

1. Introduction

The goal of this research paper is to design and evaluate an NLP system capable of determining semantic equivalence between pairs of questions. This report outlines the end-to-end development pipeline that includes data exploratory analysis, robust preprocessing methodologies, and a comparative assessment of deep learning architectures versus traditional machine learning classifiers. Specifically, a Siamese LSTM network was compared to the baseline Logistic Regression and SVM classifiers to determine the best approach toward semantic text similarity.

2. Exploratory Data Analysis (EDA)

Extensive exploratory analysis was performed to understand the statistical properties of the dataset and then pinpoint possibly predictive features.

- **Data Integrity Assessment:** First, a preliminary audit was performed to identify and mitigate missing values to ensure the reliability of the training corpus.
- **Feature Engineering:** Extracted quantitative features are the token counts of question1 and question2 to see the relationship between sentence length and semantic duplication.
- **Correlation Study:** The correlation matrix was computed and visualized in a heatmap form, quantifying the linear relationship between the derived length features and the target variable, is_duplicate.
- **Semantic Visualization:**
- **Lexical Analysis:** A Word Cloud visualization was generated for duplicate pairs, which isolated high-frequency terms and dominant topics within the positive class.
- **Class Balance:** The target variable's distribution was checked to identify if there is class imbalance.
- **Distributional Analysis:** The use of a histogram to compare the frequency distributions across question lengths helped highlight the variability of input.

3. Preprocessing Framework

A meticulous text normalization pipeline was implemented to guarantee data consistency and enhance model generalization.

Text Normalization Protocols

The following transformations were applied systematically to each input sequence:

1. **Case Normalization:** All text data was transformed into lowercase to remove case-sensitivity variations.

2. **Noise Reduction:** Regular expressions (re) were used to remove non-alphanumeric characters and punctuation.
3. **Stopword Filtration:** Removed high-frequency, low information terms, such as determinants and prepositions using the NLTK corpus to minimize noise.
4. **Stemming:** The Porter Stemming algorithm was applied, which reduced the inflected words to their root forms; for example, reducing "running" to "run". In this way, it compresses the vocabulary space and normalizes lexical variations.

4. Architectural Design and Modeling

Deep Learning Approach: Siamese LSTM

A **Siamese Neural Network** was designed to suit the particular needs of pair-based input. This network architecture is the current standard for industry semantic similarity tasks because it can learn a shared distance metric.

- **Input Specification:** The model takes dual inputs corresponding to the question pair as input.
- **Vectorization:** Text sequences were tokenized, with a vocabulary limit of 10,000, and padded to a fixed length of 30 tokens to satisfy tensor shape constraints.
- **Shared Feature Extraction:** Both the inputs pass through a shared Embedding layer, followed by a shared LSTM layer; the sharing of weights at both these layers means that both questions get mapped to a common geometric feature space using the same transformation logic.
- **Feature Fusion:** This involves concatenating the resultant context vectors from the LSTM branches into one combined representation.
- **Classification Head:**
 - **Dense Layer:** Applies non-linear transformation (ReLU activation).
 - **Regularization:** Dropout layers were integrated to mitigate overfitting, and Batch Normalization was applied to stabilize gradient descent.
 - **Output:** A final Sigmoid activation layer outputs the probability of semantic equivalence.
- **Hyperparameter Optimization:** The grid search strategy has been performed for the optimization of critical hyperparameters: LSTM Units (32, 64) and Dropout Rates (0.2, 0.3).

Machine Learning Baselines

To establish performance benchmarks, traditional discriminative models were trained utilizing **Term Frequency-Inverse Document Frequency (TF-IDF)** vectorization (5,000 features).

- **Logistic Regression:** A linear classifier optimized via maximum likelihood estimation (max iterations: 300).
- **Support Vector Machine (LinearSVC):** It is a margin-based classifier that decides the best possible hyperplane to separate duplicate and non-duplicate classes.

5. Evaluation Methodology

Model performance was rigorously evaluated on a stratified hold-out test set (20% split) utilizing the following metrics:

- **Accuracy:** The ratio of correctly classified cases with respect to the total sample size.
- **Classification Report:** A granular view of the Precision, Recall, and F1-Score for each class, giving insight into the sensitivity and specificity of the model.
- **Confusion Matrix:** To further analyze the type of misclassifications, a heatmap visualization was generated, comparing False Positives vs. False Negatives.
- **ROC-AUC Analysis:** The ROC curve and the AUC score were calculated to assess the discriminative power of the model at different decision thresholds.
- **Learning Curves:** The accuracy of training and validation versus epochs was plotted for the deep learning model to diagnose whether it has converging behavior or overfitting.

6. Conclusion

This work successfully implemented and validated a multi-paradigm NLP pipeline for duplicate question detection. The comparative analysis between the Siamese LSTM architecture and TF-IDF-based linear models provides a comprehensive view of the trade-offs between computational complexity and predictive accuracy. This will form the basis for selecting the right deployment model, considering resource constraints versus performance requirements.